

task 2: Sensor Data Simulation

const int tempPin=A0; // TMP36 signal pin

const int potPin=A1; //Potentiometer(simulated humidity sensor)

void setup() {

 serial.begin(9600);

}

void loop(){

 //Read raw sensor values

 int rawTemp=analogRead(tempPin);

 int rawhumidity=analogRead(potPin);

 //convert TMP36 raw value to actual temperature in celsius

 float voltage =rawTemp*(5.0/1023.0);

 float temperatureC=(voltage-0.5)*100.0;

 //convert potentiometer raw value (0-1023) to a 0-100%humidity scale

 float humidityPercent=map(rawHumidity,0,1023,0,100);

 //log the data

 Serial.print("Temperature:");

 Serial.print(temperatureC);

 Serial.println("C");

 Serial.print("Humidity:");

 Serial.print(humidityPercent);

 Serial.println("%");

 Serial.println("_____");

 delay(2000); //log every 2 seconds

}